



National University of Sciences and Technology (NUST)
School of Electrical Engineering and Computer Science

Department of Computing

CS220: Database Systems

Class: BSAI – 2A

Lab 09: Subqueries

CLO-2: Formulate SQL queries to retrieve information from a relational database.

Date: 09 April 2026

Time: 10:00 - 13:00



Lab 09: SQL Queries (Subqueries)

Introduction

The sub query in SQL is a query within query. A sub- query is composed of an outer query and an inner query.

Objectives

After completing this lab, you should be able to do the following:

- Understand and practice the single row sub queries
- Understand and practice the multiple row sub queries

Tools/Software Requirement

- MySQL Community Server
- MySQL Workbench

Helping Material

<https://www.w3resource.com/sql/subqueries/understanding-sql-subqueries.php>

Description:

For example, you want to see all the employees whose salary is above average salary. For this you have to first compute the average salary using AVG function and then compare employees' salaries with this computed salary. This is possible using subquery. Here the sub query will first compute the average salary and then main query will execute.

```
Select *  
from emp  
where sal >  
      (select avg(sal)  
       from emp);
```

Similarly we want to see the name and empno of that employee whose salary is maximum.

```
Select *  
from emp
```



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```
where sal = (select max(sal)
            from emp);
```

We want to see how many employees are there whose salary is above average.

```
Select count(*)
from emp
where sal >
      (select avg(sal)
       from emp);
```

We want to see those employees who are working in 'DALLAS'. Remember emp and dept are joined on deptno and loc column is in the dept table. Assuming that wherever the department is located the employee is working in that city.

```
Select *
from emp

where deptno IN
      (select deptno
       from dept
       where loc='DALLAS');
```

Lab Tasks

On the basis of the knowledge of the sub-queries, attempt the following questions using the Subqueries concept. The information needs can also be met through join operations; however, you need to practice the subqueries. Try to implement subqueries concept as much as possible. You can use join operation in your queries if required.

Write equivalent join queries for at least two queries.

Write SQL queries for the following information needs using subquery approach.

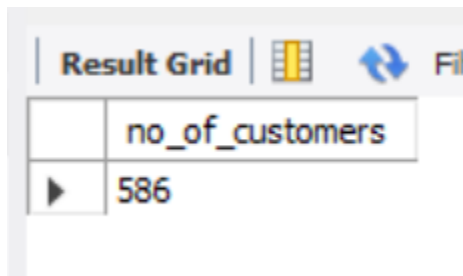
- Number of customers who never rented BREAKING HOME.



CODE:

```
Select count(*) as no_of_customers  
From customer  
where customer_id NOT IN(  
select r.customer_id  
From rental r  
Join inventory i ON i.inventory_id=r.inventory_id  
Join film f ON f.film_id=i.film_id  
where f.title="Breaking Home")
```

OUTPUT:



	no_of_customers
▶	586

- b) List accumulative replacement cost of the movies that were never rented.

CODE:

```
Select sum(replacement_cost)  
From film  
WHERE film_id NOT IN (  
SELECT i.film_id
```



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FROM inventory i

JOIN rental r ON i.inventory_id = r.inventory_id

)

OUTPUT:

Result Grid		Filter Rows:
	sum(replacement_cost)	
▶	766.58	

c) Find customers who are associated to City Abu Dhabi or Aden.

CODE:

Select first_name,last_name

From customer

where address_id IN(

Select address_id

From address

Where city_id IN(

Select city_id

From city

Where city IN("Abu Dhabi","Aden")

))



OUTPUT:

	first_name	last_name
▶	TOM	MILNER
	ELLA	OLIVER

d) What category does the movie COMA HEAD belong to?

CODE:

```
SELECT c.name
FROM category c
JOIN film_category fc ON c.category_id = fc.category_id
JOIN film f ON fc.film_id = f.film_id
WHERE f.title = 'COMA HEAD';
```

OUTPUT:

	name
▶	Travel

e) Find the movies whose rental duration is less than the rental duration of movie 'AFRICAN EGG'.



CODE:

```
SELECT title
```

```
From film
```

```
Where rental_duration<(
```

```
Select rental_duration
```

```
From film
```

```
Where title="African Egg")
```

OUTPUT:

Result Grid			Filter Rows:
	title		
▶	ACE GOLDFINGER		
	AFFAIR PREJUDICE		
	AGENT TRUMAN		
	ALABAMA DEVIL		
	ALI FOREVER		
	ALIEN CENTER		
	ALONE TRIP		

f) Find those films whose category id is greater than every category id of available films.

CODE:

```
SELECT film_id, title
```

```
FROM film f
```



WHERE EXISTS (

SELECT 1

FROM film_category fc

WHERE f.film_id = fc.film_id

AND fc.category_id > (

SELECT MAX(category_id)

FROM film_category fc2

WHERE EXISTS (

SELECT 1

FROM inventory i

WHERE fc2.film_id = i.film_id

)))

OUTPUT:

Result Grid		
	film_id	title
*	NULL	NULL

Deliverables

Save all queries and their results in the word document that you are executing along with the snapshots of the results. Upload this document to LMS.